Contents

1 Introduction **1**
1.1 Empirical dynamics under boundary queries 2

2 Finite-lag objects as boundary queries **3**
2.1 Edge regularities as primary finite-lag objects 3
2.2 Multi-time regularities 4
2.3 Kernel representations (optional) 4
2.4 Markov completion (derived) 4

3 Boundary horizons and query-relative compatibility **4**
3.1 Boundary primacy and non-privileging 4
3.2 Boundary queries 5
3.3 Query-relative compatibility 5

4 Boundary completion across resolutions **5**
4.1 Boundary queries and derived regularities 5
4.2 Projective compatibility and boundary completion 6
4.3 Exact boundary completion 7
4.4 Approximate completion 7
4.5 Derived dynamics and earned structure 7

5 Lag queries, sufficiency, and dominance **8**
5.1 Finite-lag boundary queries 8
5.2 Lag sufficiency 8
5.3 Relation to predictive conditioning (optional) 8

6 Extension and boundary completion **9**
6.1 Extension as joint realizability 9
6.2 Approximate boundary completion 9

7 Admissible dynamical structure **10**
7.1 From extension to kernel representations 10
7.2 Approximate semigroup structure 10
7.3 Generators as closure limits 11
7.4 Empirical adequacy versus explanatory realization 11

8 Information geometry on boundary queries **11**

9 Underdetermination and empirical stance **11**

1 Introduction

We work throughout on standard Borel spaces, ensuring the existence of regular conditional probabilities and allowing conditional distributions to be represented as Markov kernels when such disintegrations exist.

1.1 Empirical dynamics under boundary queries

Observable regularities and query families. We begin from a class $\mathcal{Q}$ of *boundary queries*. A boundary query is an operational procedure that extracts observable structure from an underlying stream of observations. Formally, each query $Q \in \mathcal{Q}$ is associated with an outcome space $\mathcal{O}_Q$ and is understood as a measurable projection, functional, or disintegration acting on the observable stream.

For each query $Q \in \mathcal{Q}$, we specify a set of admissible regularities

$$\mathcal{R}_Q \subseteq \mathcal{P}(\mathcal{O}_Q),$$

representing the class of empirically stable statistical structures that may arise under that query.

A *boundary horizon (query-relative)* is a family

$$\{\nu_Q\}_{Q\in\mathcal{Q}}, \qquad \nu_Q \in \mathcal{R}_Q,$$

interpreted as the empirically observed collection of regularities across all admissible queries.

Completion as an optional representation. When the observable family $\{\nu_Q\}_{Q\in\mathcal{Q}}$ is jointly realizable, there exists a path-space measure $\mathbb{P}_Y \in \mathcal{P}(Y^{\mathbb{Z}})$ such that

$$\nu_Q = Q_{\#}\mathbb{P}_Y \qquad \text{for all } Q \in \mathcal{Q}.$$

In this case, $\mathbb{P}_Y$ provides a *completion* or *process-level representation* of the observable horizon. Importantly, the existence of such a completion is not assumed. Instead, it is treated as an empirical and mathematical question.

Thus, throughout, the observable family $\{\nu_Q\}$ is taken as primitive, while path laws are regarded as justified when coherence across queries permits.

Justified dynamical structure. Finite-resolution and directional dynamical objects (regularities, conditional kernels, operators) arise from statistical interrogation of the boundary horizon. Concretely, such objects are obtained through projection, conditioning, or coarse-graining relative to auxiliary choices (such as a lag Δt or a conditioning σ-field). In this sense, dynamical descriptions are always relative to a querying regime.

When a completion $\mathbb{P}_Y$ exists, these objects may be realized concretely as measurable operations on the path space $Y^{\mathbb{Z}}$ evaluated under $\mathbb{P}_Y$. Otherwise, they are understood operationally through their action on the observable family $\{\nu_Q\}$.

Stationarity and scope. Stationarity is invoked only when global index-shift invariance or semigroup structure is required. The notions of boundary queries, compatibility, and (approximate) boundary completion depend only on local extendability of the observable regularities $\{\nu_Q\}$ across resolutions. Failure of stationarity does not preclude meaningful multi-resolution structure, but it may obstruct the identification of a single time-homogeneous generator.

Failure of (exact or approximate) boundary completion is itself a meaningful empirical outcome. Such failure indicates that the observable regularities do not admit a coherent multi-resolution closure and therefore identifies regimes in which new hypotheses or refined queries may be required.

Boundary queries and query-relative compatibility. A boundary query acts directly on the observable horizon. When a completion exists, it may be realized as a measurable map or Markov kernel

$$Q : Y^{\mathbb{Z}} \rightsquigarrow \mathcal{O}_Q,$$

inducing the law $Q_{\#}\mathbb{P}_Y$. More generally, compatibility is defined operationally.

Let $\mathcal{Q}$ be a fixed class of admissible queries. A representation s is said to be $\mathcal{Q}$-*compatible* if each observable query can be reproduced after passing through the representation:

$$s \text{ is } \mathcal{Q}\text{-compatible} \quad \Longleftrightarrow \quad \forall Q \in \mathcal{Q}\ \exists \widetilde{Q} \text{ such that } Q = \widetilde{Q} \circ s.$$

This expresses the idea that the representation preserves all observable structure specified by the query class.

Finite-lag and structural queries. No single query is privileged. Instead, we consider families of finite-resolution and structural queries adapted to the empirical context.

A basic example consists of finite-lag projections. For each $\Delta t \in H$, define

$$Q_{\Delta t} : Y^{\mathbb{Z}} \to Y \times Y, \qquad Q_{\Delta t}(y) = (y_0, y_{\Delta t}),$$

which induce the corresponding lag-regularities

$$\Gamma_Y^{(\Delta t)} := (Q_{\Delta t})_{\#}\mathbb{P}_Y$$

when a completion exists. These summaries serve as the primary objects for cross-resolution comparison and extension.

More general query families may include multi-time projections, coarse-grainings, or experimentally defined observables. Compatibility and completion are therefore always understood relative to the chosen class $\mathcal{Q}$.

Conditional and predictive structure as derived queries. Conditional or predictive descriptions arise as special cases when suitable disintegrations exist. For example, conditioning future observations on the past yields predictive kernels. Within the present framework, such structures are treated as derived statistical queries rather than as primitive dynamical assumptions. Their relevance depends on the empirical setting and the class of admissible queries.

2 Finite-lag objects as boundary queries

We now introduce the basic observable summaries obtained by applying finite-resolution queries to the empirical horizon. These summaries form the primary objects used for comparison across observational scales.

2.1 Edge regularities as primary finite-lag objects

Let Y be a standard Borel space Y, and let $\pi_t : Y^{\mathbb{Z}} \to Y$ denote the coordinate projection at time t.

Definition 1 (Edge regularity)**.** *For any lag $\Delta t \in H$, the* edge regularity *at scale Δt is the two-time regularity $\Gamma_Y^{(\Delta t)} \in \mathcal{P}(Y \times Y)$ induced by the lag query $Q_{\Delta t}(y) = (y_0, y_{\Delta t})$. When a path-law completion $\mathbb{P}_Y \in \mathcal{P}(Y^{\mathbb{Z}})$ exists, this is the pushforward $(\pi_0, \pi_{\Delta t})_{\#}\mathbb{P}_Y$.*

We interpret $\Gamma_Y^{(\Delta t)}$ as the outcome of a *finite-lag query*—an observational procedure, not a dynamical law. Different choices of Δt correspond to different resolutions at which the empirical horizon is interrogated.

When marginal coherence holds across the family $\{\Gamma_Y^{(\Delta t)}\}$ (see Section 4), we denote the common one-time marginal by

$$\mu_Y \in \mathcal{P}(Y),$$

and note that $\mu_Y = (\pi_0)_\# \mathbb{P}_Y$ when a completion exists.

2.2 Multi-time regularities

More detailed summaries arise from projections onto multiple coordinates.

Definition 2 (Three-time regularity)**.** *For lags $\Delta t_1, \Delta t_2 \in H$, define the three-time regularity*

$$\Gamma_Y^{(\Delta t_1, \Delta t_2)} := (\pi_0, \pi_{\Delta t_1}, \pi_{\Delta t_1 + \Delta t_2})_\# \, \mathbb{P}_Y \;\in\; \mathcal{P}(Y^3).$$

These summaries capture observable relationships across multiple scales and will play a central role in assessing cross-resolution coherence.

2.3 Kernel representations (optional)

Whenever an edge regularity admits a disintegration, one may introduce a kernel representation.

Definition 3 (Kernel representation)**.** *If*

$$\Gamma_Y^{(\Delta t)}(dy, dy') = \mu_Y(dy)\, T_{\Delta t}(y, dy'),$$

then $T_{\Delta t} : Y \rightsquigarrow Y$ is called a kernel representation of the lag-Δt regularity.

Such representations provide convenient operator forms but are not required. Given a kernel, one obtains a linear operator on observables $f : Y \to \mathbb{R}$:

$$(T_{\Delta t} f)(y) := \int_Y f(y')\, T_{\Delta t}(y, dy').$$

2.4 Markov completion (derived)

A family of kernels $\{T_{\Delta t}\}_{\Delta t \in H}$ is said to form a Markov completion when it approximately closes across lags:

$$T_{\Delta t_1 + \Delta t_2} \;\approx\; T_{\Delta t_1} \circ T_{\Delta t_2},$$

with approximation quantified by a divergence criterion on the induced edge regularities.

Continuity in Δt may then permit the introduction of a generator as a limiting representation.

3 Boundary horizons and query-relative compatibility

3.1 Boundary primacy and non-privileging

Proposition 1 (Boundary primacy)**.** *Let Y be a standard Borel space and let $\mathbb{P}_Y \in \mathcal{P}(Y^{\mathbb{Z}})$ denote a boundary horizon. All finite-resolution and directional structures (e.g. regularities, kernels, operators) arise relative to a boundary query. No finite resolution is intrinsically distinguished.*

Proof. This follows from the operational construction of observable structure introduced in Section 1. □

Remark 1 (Non-privileging of resolution)**.** *No canonical temporal discretization, lag, or compositional scale is assumed. Observable summaries depend on the chosen querying regime.*

3.2 Boundary queries

Definition 4 (Boundary query)**.** *A boundary query is any measurable operation on the path space realized by projection, disintegration, or composition thereof. Equivalently, it may be represented as a Markov kernel*

$$Q : Y^{\mathbb{Z}} \rightsquigarrow \mathcal{O},$$

inducing a regularity $Q_{\#}\mathbb{P}_Y$.

3.3 Query-relative compatibility

Let $\mathcal{Q}$ be a class of admissible queries.

Definition 5 (Compatibility)**.** *A representation*

$$s : Y^{\mathbb{Z}} \to \mathcal{S}$$

is $\mathcal{Q}$*-compatible if for every* $Q \in \mathcal{Q}$ *there exists* $\widetilde{Q}$ *such that*

$$Q = \widetilde{Q} \circ s.$$

Proposition 2 (Universal property)**.** *A representation is compatible if and only if all queries in* $\mathcal{Q}$ *factor through it.*

Proof. Immediate from Definition 5. □

Remark 2. *Compatibility is relative to* $\mathcal{Q}$*; distinct compatible representations may differ substantially in internal structure.*

4 Boundary completion across resolutions

We now study when observable regularities at different resolutions exhibit coherent cross-scale structure. The notion of *boundary completion* is empirical and pre-dynamical: it concerns joint extendability (consistency) of query-induced regularities across resolutions, and does not assume any underlying evolution law.

4.1 Boundary queries and derived regularities

Fix a set of lags $H \subset \mathbb{R}_+$ (or $H \subset \mathbb{Z}_+$). For each $\Delta t \in H$, consider the finite-lag query

$$Q_{\Delta t} : Y^{\mathbb{Z}} \to Y \times Y, \qquad Q_{\Delta t}(y) = (y_0, y_{\Delta t}).$$

A query-relative horizon specifies the induced two-time regularity at lag Δt as

$$\Gamma_Y^{(\Delta t)} \in \mathcal{P}(Y \times Y).$$

When a path-law completion $\mathbb{P}_Y \in \mathcal{P}(Y^{\mathbb{Z}})$ exists and $Q_{\Delta t}$ is realized as a measurable projection, this agrees with the pushforward $\Gamma_Y^{(\Delta t)} = (Q_{\Delta t})_{\#}\mathbb{P}_Y$.

We write

$$\mathcal{B}_Y(H) := \{\Gamma_Y^{(\Delta t)} : \Delta t \in H\} \subset \mathcal{P}(Y \times Y)$$

for the family of lag-H boundary regularities.

4.2 Projective compatibility and boundary completion

We now make precise the relationship between query-relative boundary regularities and classical consistency conditions for stochastic processes. The key point is that the boundary framework generalizes Kolmogorov consistency from coordinate projections to arbitrary classes of observational queries.

Query systems as projective systems. Let $\mathcal{Q}$ be a class of boundary queries, each associated with a standard Borel outcome space $\mathcal{O}_Q$. Assume $\mathcal{Q}$ is equipped with a preorder $\preceq$ expressing refinement: for $Q_1, Q_2 \in \mathcal{Q}$, we write $Q_1 \preceq Q_2$ if Q_2 refines Q_1, meaning that there exists a measurable map

$$\pi_{21} : \mathcal{O}_{Q_2} \to \mathcal{O}_{Q_1}$$

such that

$$Q_1 = \pi_{21} \circ Q_2.$$

Thus the query class induces a projective system of measurable spaces.

Projective families of boundary regularities. A family of boundary regularities

$$\{\nu_Q\}_{Q\in\mathcal{Q}}, \qquad \nu_Q \in \mathcal{P}(\mathcal{O}_Q),$$

is said to be *projectively compatible* if for all $Q_1 \preceq Q_2$,

$$\nu_{Q_1} = (\pi_{21})_{\#}\nu_{Q_2}.$$

This condition expresses exact stability of observable regularities under refinement of the querying regime.

Generalized Kolmogorov extension.

Theorem 1 (Generalized Kolmogorov extension)**.** *Let Y be a standard Borel space and let $\mathcal{Q}$ be a directed class of boundary queries with refinement preorder. Suppose that the family $\{\nu_Q\}_{Q\in\mathcal{Q}}$ is projectively compatible and that the σ-algebra generated by the query cylinders separates points in $Y^{\mathbb{Z}}$. Then there exists a probability measure*

$$\mathbb{P}_Y \in \mathcal{P}(Y^{\mathbb{Z}})$$

such that

$$\nu_Q = Q_{\#}\mathbb{P}_Y \qquad \text{for all } Q \in \mathcal{Q}.$$

Moreover, this completion is unique on the σ-algebra generated by the query family.

Remark 3 (Classical Kolmogorov as a special case)**.** *If $\mathcal{Q}$ consists of all finite coordinate projections, then projective compatibility reduces to classical Kolmogorov consistency. Thus stochastic processes arise as special cases of compatibility across coordinate queries.*

Remark 4 (Interpretation)**.** *The theorem shows that a path-space boundary horizon, when it exists, can be regarded as the projective limit of observable regularities. In this sense, global path-law structure is not assumed but arises as a completion of compatible empirical summaries.*

4.3 Exact boundary completion

Definition 6 (Exact completion)**.** *The family $\mathcal{B}_Y(H)$ is* boundary complete *if:*

1. *(*Marginal coherence*) All lag regularities share a common one-time marginal:*

$$(\pi_0)_\# \Gamma_Y^{(\Delta t)} = \mu_Y \qquad \textit{for all } \Delta t \in H,$$

   for some $\mu_Y \in \mathcal{P}(Y)$.

2. *(*Two-step extendability*) For all $\Delta t_1, \Delta t_2 \in H$ with $\Delta t_1 + \Delta t_2 \in H$, the extension set*

$$\mathrm{Ext}(\Gamma_Y^{(\Delta t_1)}, \Gamma_Y^{(\Delta t_2)})$$

   is nonempty.

This condition expresses joint extendability of observable summaries, and precedes any Markov or semigroup interpretation.

4.4 Approximate completion

Exact consistency is often too strong. We therefore quantify deviations from extendability.

Definition 7 (ε–completion)**.** *The family is ε–boundary complete if for all $\Delta t_1, \Delta t_2 \in H$ with $\Delta t_1 + \Delta t_2 \in H$,*

$$\inf_{\widetilde{\Gamma} \in \mathrm{Ext}(\Gamma_Y^{(\Delta t_1)}, \Gamma_Y^{(\Delta t_2)})} D_f\big(\Gamma_Y^{(\Delta t_1 + \Delta t_2)}, \widetilde{\Gamma}\big) \le \varepsilon.$$

4.5 Derived dynamics and earned structure

Boundary completion determines which additional structures may be introduced without contradicting observable regularities.

- Kernel representations arise only when edge regularities admit disintegration.
- Semigroup structure reflects cross-resolution closure of the family $\mathcal{B}_Y(H)$.
- Generators, when they exist, arise as limiting objects associated with increasingly refined observational scales.

Interior dynamical realizations are therefore admissible only insofar as they reproduce the completed boundary structure. They serve as explanatory devices rather than primitive sources of empirical regularity.

In this sense, all dynamical structure in the framework is *earned from the boundary*, rather than imposed upon it.

5 Lag queries, sufficiency, and dominance

5.1 Finite-lag boundary queries

Throughout, we treat the family of two-time boundary regularities

$$\mathcal{B}_Y(H) := \left\{\Gamma_Y^{(\Delta t)} \in \mathcal{P}(Y \times Y)\right\}_{\Delta t \in H}$$

as the primary finite-resolution empirical summaries of the boundary horizon. When a path-space regularity $\mathbb{P}_Y \in \mathcal{P}(Y^{\mathbb{Z}})$ exists, these may be realized as

$$\Gamma_Y^{(\Delta t)} = (\pi_0, \pi_{\Delta t})_{\#}\mathbb{P}_Y,$$

but we do *not* assume such a completion a priori.

We refer to each $\Gamma_Y^{(\Delta t)}$ as a *lag-Δt boundary query.* Such queries are understood as statistical observations of the boundary horizon, not as mechanisms that generate it.

Remark 5 (No privileged time step)**.** *We do not single out any distinguished "one-step" lag or canonical discretization. All $\Delta t \in H$ are treated symmetrically; structure is only admissible under relations among these queries.*

5.2 Lag sufficiency

Definition 8 (Lag sufficiency)**.** *The boundary family $\mathcal{B}_Y(H)$ is said to be* lag-sufficient *if every finite-dimensional boundary regularity that is empirically meaningful for the system is determined (up to the ambiguity of extension) by the collection $\{\Gamma_Y^{(\Delta t)}\}_{\Delta t \in H}$.*

Intuitively, lag sufficiency expresses that all higher-order finite-resolution summaries are shadows of, or constraints upon, the two-time regularities.

Proposition 3 (Lag dominance)**.** *Suppose $\mathcal{B}_Y(H)$ is exactly boundary complete (Definition 6). Then every finite-dimensional marginal of any admissible path-space completion of the boundary horizon is compatible with, and constrained by, the family $\{\Gamma_Y^{(\Delta t)}\}_{\Delta t \in H}$.*

Sketch. Exact boundary completion ensures that for any finite collection of times $t_0 < t_1 < \cdots < t_n$ with increments in H, there exists a joint law on Y^{n+1} whose adjacent marginals coincide with the corresponding edge regularities. Thus any higher-order finite-dimensional summary must be realizable via successive extensions of lag queries, and hence cannot contradict them. □

Corollary 1 (Minimality of lag sufficiency)**.** *Any representation that preserves all lag-Δt boundary regularities $\{\Gamma_Y^{(\Delta t)}\}_{\Delta t \in H}$ preserves all finite-lag empirical content of the boundary horizon.*

Remark 6 (Lag sufficiency is not Markovity)**.** *Lag sufficiency does* not *imply the existence of Markov kernels, conditional independence, or a semigroup structure. It is a statement about empirical summaries, not about dynamics.*

5.3 Relation to predictive conditioning (optional)

If a path-space regularity $\mathbb{P}_Y$ exists and admits disintegration, one may form the predictive kernel

$$K_\infty(\cdot \mid p) := \mathbb{P}_Y(Y_{0:\infty} \in \cdot \mid Y_{-\infty:0} = p).$$

In that case, each $\Gamma_Y^{(\Delta t)}$ is a marginal of K_∞.

However, we do *not* take K_∞ to be primitive or canonical. Rather, it is a derived object that may be useful in some contexts, but whose relevance is contingent on additional structure beyond lag sufficiency and boundary completion.

6 Extension and boundary completion

6.1 Extension as joint realizability

Definition 9 (Extension set). *Let $\Gamma_1, \Gamma_2 \in \mathcal{P}(Y \times Y)$ be two boundary regularities for successive lags (e.g. $\Gamma_1 = \Gamma_Y^{(\Delta t_1)}$ and $\Gamma_2 = \Gamma_Y^{(\Delta t_2)}$). Assume they share a compatible intermediate marginal in the sense that there exists $\mu \in \mathcal{P}(Y)$ such that*

$$(\pi_2)_\# \Gamma_1 = \mu = (\pi_1)_\# \Gamma_2.$$

Define the extension set

$$\mathrm{Ext}(\Gamma_1, \Gamma_2) := \Big\{ (\pi_0, \pi_2)_\# \Lambda : \Lambda \in \mathcal{P}(Y^3),\ (\pi_0, \pi_1)_\# \Lambda = \Gamma_1,\ (\pi_1, \pi_2)_\# \Lambda = \Gamma_2 \Big\}.$$

Elements of $\mathrm{Ext}(\Gamma_1, \Gamma_2)$ *are precisely the endpoint regularities that can arise from a three-time joint law consistent with the prescribed two-time regularities.*

Proposition 4 (Existence of extensions). $\mathrm{Ext}(\Gamma_1, \Gamma_2) \neq \varnothing$ *if and only if* Γ_1 *and* Γ_2 *share a common intermediate marginal* μ*.*

Sketch. If $\mathrm{Ext}(\Gamma_1, \Gamma_2) \neq \varnothing$, any witness Λ must satisfy $(\pi_2)_\# \Gamma_1 = (\pi_1)_\# \Gamma_2$ by construction. Conversely, if such a marginal μ exists, standard gluing arguments for probability measures on standard Borel spaces guarantee the existence of some Λ with the required pairwise marginals. □

Remark 7 (Extension is not composition). Ext *expresses a condition of* joint realizability *among empirical regularities. It does not assume Markov kernels, conditional independence, causal direction, or any compositional law on Y. Time order and dynamics are not built into the definition.*

6.2 Approximate boundary completion

Fix a divergence $D_f(\cdot, \cdot)$ on probability measures and a tolerance $\varepsilon > 0$.

Definition 10 (ε–boundary completion). *The boundary family $\mathcal{B}_Y(H)$ is said to be ε–*boundary complete *if for all $\Delta t_1, \Delta t_2 \in H$,*

$$\inf_{\widetilde{\Gamma} \in \mathrm{Ext}\left(\Gamma_Y^{(\Delta t_1)}, \Gamma_Y^{(\Delta t_2)}\right)} D_f\Big((\pi_0, \pi_{\Delta t_1 + \Delta t_2})_\# \mathbb{P}_Y,\ \widetilde{\Gamma} \Big) \ \le\ \varepsilon.$$

Conceptually, ε–boundary completion measures how closely the empirically observed long-lag boundary regularity can be approximated by *some* joint realization consistent with the corresponding shorter-lag regularities.

This definition refers only to joint extendability of boundary summaries and does *not* assume Markov kernels, conditional independence, or causal composition.

Remark 8 (Interpretation of ε). *The parameter ε admits three complementary readings:*

1. *As a measure of* non-extendability *across resolutions;*

2. *As a bound on* mutual expressibility *of boundary viewpoints;*

3. *As an* operational budget *for hypothesis generation when exact completion fails.*

Large ε signals genuine incompatibility of boundary summaries, not merely data noise.

7 Admissible dynamical structure

The preceding sections define boundary regularities and their cross-resolution consistency without invoking any notion of dynamics. We now describe how familiar dynamical objects—Markov kernels, semigroups, and generators—may be *admissible* as representations of boundary structure when additional regularity is present. Nothing in this section is assumed at the boundary level; all constructions are conditional and generally non-unique.

7.1 From extension to kernel representations

Proposition 5 (Admissible kernel representation)**.** *Let $\mathcal{B}_Y(H)$ be an ε–boundary complete family of boundary regularities. Suppose that for each $\Delta t \in H$, the corresponding edge regularity $\Gamma_Y^{(\Delta t)} \in \mathcal{P}(Y \times Y)$ admits a disintegration of the form*

$$\Gamma_Y^{(\Delta t)}(dy, dy') = \mu_Y(dy)\, T_{\Delta t}(y, dy'),$$

for some probability kernel $T_{\Delta t} : Y \rightsquigarrow Y$. Then $\{T_{\Delta t}\}_{\Delta t \in H}$ provides a kernel-based representation of the boundary regularities consistent with the given querying regime.

Remark 9. *The existence of kernels $T_{\Delta t}$ is not guaranteed by boundary completion alone. Disintegration is an additional structural property of the boundary regularities, and kernel representations are introduced only when such disintegrations exist.*

Remark 10 (Non-uniqueness of kernel representations)**.** *Even when disintegrations exist, kernel representations are generally non-unique. Multiple families of kernels may induce the same boundary regularities and hence be equally compatible with the boundary horizon. This reflects underdetermination at the level of interior dynamics.*

7.2 Approximate semigroup structure

Proposition 6 (Approximate semigroup emergence)**.** *Suppose that $\mathcal{B}_Y(H)$ is ε–boundary complete and admits kernel representations $\{T_{\Delta t}\}_{\Delta t \in H}$. Then for $\Delta t_1, \Delta t_2 \in H$, the composed kernel $T_{\Delta t_1} \circ T_{\Delta t_2}$ approximates $T_{\Delta t_1 + \Delta t_2}$ in the sense that*

$$D_f\Big(\mu_Y \otimes (T_{\Delta t_1} \circ T_{\Delta t_2}),\ \Gamma_Y^{(\Delta t_1 + \Delta t_2)}\Big) \lesssim \varepsilon,$$

where the comparison is mediated by the boundary completion tolerance.

Sketch. By ε–boundary completion, there exists some $\widetilde{\Gamma} \in \mathrm{Ext}(\Gamma_Y^{(\Delta t_1)}, \Gamma_Y^{(\Delta t_2)})$ such that

$$D_f\Big(\Gamma_Y^{(\Delta t_1 + \Delta t_2)}, \widetilde{\Gamma}\Big) \le \varepsilon.$$

When disintegrations exist, the product-form endpoint law $\mu_Y \otimes (T_{\Delta t_1} \circ T_{\Delta t_2})$ is an admissible element of the same extension set. Selecting this witness yields the stated approximation. □

Remark 11. *The approximate semigroup property is not imposed but follows from the consistency of boundary regularities across resolutions. The quality of the approximation is controlled by ε and may vary across scales.*

7.3 Generators as closure limits

Theorem 2 (Generator as closure limit)**.** *Assume that the family $\{T_{\Delta t}\}_{\Delta t \in H}$ admits a limit in an appropriate operator topology as $\Delta t \to 0$. When such a limit exists, it defines an infinitesimal generator L that provides a continuous-time dynamical representation of the boundary regularities.*

Remark 12 (Conditional status of generators)**.** *The existence, uniqueness, and interpretation of generators depend on additional analytical assumptions beyond boundary completion, including continuity, topology, and limiting behavior. Generators are therefore optional representational tools rather than mandatory consequences of the framework.*

7.4 Empirical adequacy versus explanatory realization

Remark 13 (Boundary adequacy and explanation)**.** *Boundary compatibility and completion constrain empirical adequacy. Kernel-based dynamics and generators provide explanatory structure but are not uniquely determined by boundary data. The framework therefore separates empirical regularity (boundary adequacy) from explanatory commitment (interior realization).*

Remark 14 (Over- and under-explanation)**.** *A realization may be empirically adequate yet introduce excess structure not warranted by the boundary horizon, or conversely may fail to capture boundary regularities despite rich internal dynamics. Such distinctions become visible only after separating boundary completion from dynamical representation.*

8 Information geometry on boundary queries

Proposition 7 (Query-induced geometry)**.**

Theorem 3 (Invariance under sufficient statistics)**.**

Remark 15 (Divergence choice and commensurability)**.**

9 Underdetermination and empirical stance

Proposition 8 (Multiplicity of compatible realizations)**.**

Remark 16 (Empirical adequacy vs explanation)**.**

Remark 17 (Failure of completion as signal)**.**

Reflection questions

The following questions are intended to audit the conceptual stance of the framework and to identify remaining points of pressure or underdetermination. They are not requirements imposed by the formalism, but diagnostics for its interpretation and scope.

Boundary completion and empirical meaning.

1. Does the definition of exact boundary completion successfully characterize *empirical coherence* among finite-resolution regularities without presupposing any underlying generative or causal process?

2. Can failure of boundary completion be an informative empirical outcome—i.e. does it distinguish genuine incompatibility of regularities from insufficient data or estimation error?

3. Is approximate boundary completion ($\varepsilon > 0$) clearly interpreted as a graded empirical property of observed regularities, rather than as a defect to be repaired by modeling?

Extension versus dynamics.

1. Is the notion of extension (Ext) unambiguously understood as a condition of *joint realizability*, rather than as an implicit form of temporal composition or causal evolution?

2. After Sections 6 and 7, is it clear that kernel composition and semigroup structure are representational conveniences that may be admissible, rather than assumptions imposed at the boundary level?

3. Would a reader unfamiliar with Markov processes nonetheless understand extension as a purely probabilistic constraint, independent of time direction or mechanism?

Admissible structure and underdetermination.

1. Is it explicit—and acceptable within the framework—that multiple, non-equivalent kernel families or generators may be equally compatible with the same boundary regularities?

2. Is the separation between *boundary adequacy* (empirical compatibility) and *explanatory realization* (interior dynamical structure) sufficiently sharp to prevent conflating predictive success with causal explanation?

3. When multiple realizations are boundary-compatible, does the framework leave their selection as an external (pragmatic or scientific) choice rather than an internal requirement?

Role and interpretation of ε.

1. Should ε be interpreted primarily as a tolerance for empirical noise, a bound on cross-resolution expressibility, or an operational budget for hypothesis generation? Is room left for all three interpretations?

2. Are there plausible scenarios in which ε is small locally (at some resolutions) but large globally, and does the framework accommodate such scale-dependent behavior?

Scope and limits of the stance.

1. Does the framework remain coherent if no boundary horizon exists even as an idealization, yet approximate completion holds at all empirically accessible resolutions?

2. Is it clear which claims in the manuscript are empirical (about observed regularities), structural (about joint realizability), and representational (about kernels, semigroups, or generators)?

3. Finally, is this a stance one is willing to be dogmatic about—namely, that all dynamical structure must be admissible from boundary regularities and never assumed in advance?